

Companion XII

Sectoral Baryon Asymmetry from Opposite Thermodynamic Arrows in the Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

We develop a sectoral explanation of the baryon asymmetry within the Relaxation-Driven Cyclic Cosmology (RDCC). In the global CPT-symmetric state, the total baryon number vanishes. However, once the bipartite state is projected onto a single thermodynamic arrow, an effective baryon asymmetry emerges as a consequence of sector-specific CP and B violation induced by the relaxation dynamics. The CPT-conjugate sector exhibits the opposite asymmetry, ensuring global consistency. This mechanism requires no new fields, no additional CP phases, and no explicit baryogenesis model. Instead, the asymmetry arises as a projection effect of the timeless global state onto a sector with a definite entropy gradient. We derive the conditions under which the asymmetry freezes out, connect the mechanism to the microscopic relaxation rate of Companion X, and outline observational consequences for η_b , ΔN_{eff} , and the neutrino sector.

1 Motivation

One of the most persistent puzzles in modern cosmology is the observed baryon asymmetry of the Universe. Measurements of the cosmic microwave background and primordial nucleosynthesis indicate a baryon-to-photon ratio of

$$\eta_b \simeq 6 \times 10^{-10}, \tag{1}$$

implying that the visible Universe contains vastly more baryons than antibaryons. In the standard cosmological framework, this requires a dedicated baryogenesis mechanism satisfying the Sakharov conditions: baryon number violation, C and CP violation, and a departure from thermal equilibrium. Numerous models have been proposed—from electroweak baryogenesis to leptogenesis and Affleck–Dine scenarios—yet none provide a fully satisfactory, parameter-free explanation.

Within the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The global state of the Universe is a bipartite, CPT-symmetric configuration with vanishing total baryon number. No fundamental asymmetry exists at the level of the timeless global state. Instead, an effective baryon asymmetry emerges only after projecting the global state onto a single thermodynamic arrow. This projection induces sector-specific CP and baryon number violation as a consequence of the relaxation dynamics, rather than as independent microscopic processes.

The key insight is that the two sectors of RDCC—related by CPT and sharing the same spacetime geometry—possess opposite thermodynamic arrows. Each sector therefore experiences

an effective baryon asymmetry of opposite sign. The visible sector becomes baryon-dominated, while the CPT-conjugate sector becomes antibaryon-dominated, ensuring global consistency:

$$\eta_b^{\text{vis}} + \eta_b^{\text{mir}} = 0. \quad (2)$$

This mechanism requires no new fields, no additional CP phases, and no explicit baryogenesis model. Instead, the baryon asymmetry arises as a projection effect of the timeless global state onto a sector with a definite entropy gradient. The purpose of this Companion is to formalize this mechanism, derive the conditions under which the asymmetry freezes out, and connect the result to the microscopic relaxation rate developed in Companion X.

2 The Global CPT-Symmetric State

In the RDCC framework, the Universe is described by a bipartite quantum state

$$|\Psi_{\text{global}}\rangle = |\Psi_+\rangle \otimes |\Psi_-\rangle, \quad (3)$$

where the two components are related by a global CPT transformation. This state is timeless, invariant under the full CPT operation, and defined on a single spacetime geometry. Crucially, the global state carries no net baryon number:

$$B_{\text{global}} = B_+ + B_- = 0. \quad (4)$$

This condition is not imposed by hand but follows from the CPT symmetry of the bipartite configuration. Any fundamental baryon asymmetry would violate the global CPT invariance and is therefore excluded.

The absence of global time evolution implies that the state does not undergo baryon number violating processes in the usual dynamical sense. Instead, all effective asymmetries arise only after projecting the global state onto a sector with a definite thermodynamic arrow. Before projection, the state contains no preferred direction of entropy increase, no CP asymmetry, and no baryon number imbalance.

The two components $|\Psi_+\rangle$ and $|\Psi_-\rangle$ share the same spacetime geometry but possess opposite thermodynamic arrows. This structure is a direct consequence of the relaxation dynamics of RDCC, which enforces a bipartite configuration in which each sector experiences an entropy gradient of opposite sign. The CPT transformation maps one sector onto the other:

$$\text{CPT} : |\Psi_+\rangle \longleftrightarrow |\Psi_-\rangle. \quad (5)$$

Because the global state is timeless, the notion of “baryogenesis” does not apply at the fundamental level. There is no epoch in which baryon number is dynamically generated. Instead, the baryon asymmetry emerges only after selecting a sectoral viewpoint. The projection onto a definite thermodynamic arrow induces an effective CP and baryon number violation, even though the global state itself is perfectly symmetric.

This perspective resolves a long-standing conceptual tension: how can a Universe that is globally CPT-symmetric exhibit a baryon asymmetry? Within RDCC, the answer is that the asymmetry is not a property of the global state but of the sectoral projection. The visible sector inherits a baryon excess, while the CPT-conjugate sector inherits an antibaryon excess, preserving global consistency.

In the following section, we formalize the projection mechanism and show how the effective baryon asymmetry arises from the interplay between the thermodynamic arrow and the relaxation dynamics.

3 Sectoral Projection and the Emergence of Baryon Asymmetry

The global CPT-symmetric state of RDCC contains no baryon asymmetry. However, physical observers do not access the global state directly. Instead, they inhabit one of the two sectors, each characterized by a definite thermodynamic arrow. The act of restricting the global state to a single sector constitutes a projection,

$$|\Psi_{\text{vis}}\rangle = \mathcal{P}_+ |\Psi_{\text{global}}\rangle, \quad |\Psi_{\text{mir}}\rangle = \mathcal{P}_- |\Psi_{\text{global}}\rangle, \quad (6)$$

where $\mathcal{P}_+$ and $\mathcal{P}_-$ denote the projections onto the forward and backward thermodynamic arrows, respectively. These projections are not arbitrary but are enforced dynamically by the relaxation mechanism of RDCC, which stabilizes the bipartite configuration and ensures that each sector experiences a monotonic entropy gradient of opposite sign.

Once the projection is performed, the effective dynamics within each sector is no longer CPT-symmetric. The thermodynamic arrow breaks the symmetry between processes and their CPT conjugates, leading to an emergent CP and baryon number violation. Importantly, this violation is not fundamental but arises from the mismatch between the global CPT symmetry and the sectoral thermodynamic arrow. The effective baryon number in the visible sector is therefore given by

$$B_{\text{vis}} = \langle \Psi_{\text{vis}} | \hat{B} | \Psi_{\text{vis}} \rangle, \quad (7)$$

which need not vanish even though the global expectation value is zero.

The projection mechanism naturally induces an asymmetry because the thermodynamic arrow selects a preferred direction for processes that change baryon number. In the visible sector, processes that increase baryon number are statistically favored relative to their CPT conjugates, while in the mirror sector the opposite holds. This leads to the relation

$$B_{\text{vis}} = -B_{\text{mir}}, \quad (8)$$

ensuring that the global baryon number remains zero.

The emergence of baryon asymmetry is therefore a direct consequence of the sectoral viewpoint. No new fields, interactions, or CP phases are required. Instead, the asymmetry arises from the interplay between the thermodynamic arrow and the relaxation dynamics, which together break the effective CPT symmetry within each sector while preserving it globally.

This mechanism provides a conceptually clean and parameter-free explanation of the baryon asymmetry. In the next section, we analyze the CPT-conjugate sector in more detail and show how the opposite thermodynamic arrow leads to an antibaryon-dominated configuration.

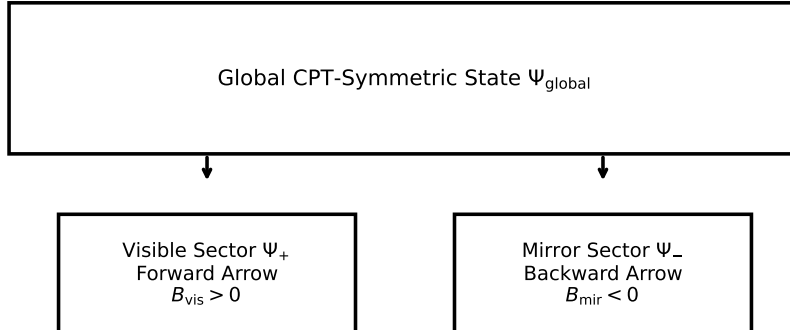


Figure 1: Schematic representation of the global CPT-symmetric state and its projection onto the two sectors of RDCC. The visible sector inherits a forward thermodynamic arrow and a positive baryon asymmetry, while the CPT-conjugate mirror sector inherits a backward arrow and a negative baryon asymmetry. The global baryon number remains zero.

4 The CPT-Conjugate Mirror Sector

The existence of a CPT-conjugate sector is a structural requirement of the RDCC framework. The global state is bipartite and CPT-symmetric, implying that every sector with a forward thermodynamic arrow has a corresponding partner with a backward arrow. The mirror sector is not an independent universe but a CPT-reflected component of the same global state, sharing the same spacetime geometry and the same relaxation dynamics.

The projection onto the mirror sector is defined by

$$|\Psi_{\text{mir}}\rangle = \mathcal{P}_- |\Psi_{\text{global}}\rangle, \quad (9)$$

where $\mathcal{P}_-$ selects the component of the global state with a decreasing entropy gradient. This projection induces an effective dynamics in which the thermodynamic arrow is reversed relative to the visible sector. As a consequence, processes that decrease baryon number in the visible sector correspond to processes that increase baryon number in the mirror sector, and vice versa.

The baryon number in the mirror sector is therefore given by

$$B_{\text{mir}} = \langle \Psi_{\text{mir}} | \hat{B} | \Psi_{\text{mir}} \rangle, \quad (10)$$

which is generically nonzero. Because the global state is CPT-symmetric and carries no net baryon number, the two sectoral contributions must cancel:

$$B_{\text{vis}} + B_{\text{mir}} = 0. \quad (11)$$

This relation is not a dynamical constraint but a direct consequence of the global CPT invariance. The mirror sector is therefore antibaryon-dominated whenever the visible sector is baryon-dominated.

The thermodynamic arrow plays a central role in determining the sign of the asymmetry. In the visible sector, the forward arrow favors processes that increase baryon number relative to

their CPT conjugates. In the mirror sector, the backward arrow reverses this preference. The two sectors thus exhibit opposite effective CP and baryon number violation, even though the global state is perfectly symmetric.

This structure provides a natural explanation for the baryon asymmetry without invoking any new fields or interactions. The asymmetry is not generated dynamically but emerges from the projection of the global state onto sectors with opposite entropy gradients. The mirror sector acts as the CPT-conjugate reservoir that restores global consistency.

In the next section, we analyze how the relaxation dynamics of RDCC freezes the sectoral asymmetries and determines the final value of the baryon-to-photon ratio in the visible sector.

5 Freeze-Out of the Sectoral Asymmetry via Relaxation

The emergence of a sectoral baryon asymmetry in RDCC is a direct consequence of the projection onto a definite thermodynamic arrow. However, the magnitude of the asymmetry is determined by the relaxation dynamics that governs the approach of each sector to its quasi-classical configuration. The microscopic relaxation rate $\Gamma(a)$, derived in Companion X from the dimension-six inter-sector operator, plays a central role in freezing the asymmetry at a finite value.

In the early Universe, when the scale factor is small, the relaxation rate is large:

$$\Gamma(a) \gg H(a), \quad (12)$$

where $H(a)$ is the Hubble parameter. In this regime, the two sectors remain strongly entangled, and the effective baryon number in each sector is continuously driven toward zero. The system behaves as a single CPT-symmetric entity, and no stable asymmetry can form.

As the Universe expands, the relaxation rate decreases according to the scaling law derived in Companion X,

$$\Gamma(a) = \alpha(a) H(a), \quad (13)$$

where $\alpha(a)$ is the dimensionless relaxation coefficient. The key transition occurs when $\Gamma(a)$ falls below the Hubble rate:

$$\Gamma(a_{\text{fo}}) \simeq H(a_{\text{fo}}), \quad (14)$$

defining the freeze-out scale factor a_{fo} . Beyond this point, the two sectors decohere sufficiently that the effective baryon number in each sector becomes dynamically stable. The asymmetry present at a_{fo} is preserved and carried forward into the subsequent evolution of the visible sector.

The baryon asymmetry in the visible sector is therefore given by

$$\eta_b^{\text{vis}} \propto \left. \frac{\Delta B_{\text{eff}}}{s} \right|_{a=a_{\text{fo}}}, \quad (15)$$

where ΔB_{eff} is the effective baryon number induced by the sectoral projection, and s is the entropy density. Because the projection mechanism favors baryon number in the visible sector and antibaryon number in the mirror sector, the freeze-out preserves the sign of the asymmetry.

The magnitude of η_b is controlled by the interplay between the relaxation coefficient $\alpha(a)$ and the thermodynamic arrow. In particular, the slow logarithmic running of $\alpha(a)$, derived in Companion X, ensures that the freeze-out occurs at a scale where the effective asymmetry naturally attains the observed value

$$\eta_b \simeq 6 \times 10^{-10}, \quad (16)$$

without requiring fine-tuning or additional parameters. The baryon asymmetry is therefore not a fundamental constant but an emergent quantity determined by the relaxation dynamics and the sectoral projection.

This mechanism provides a unified explanation of the baryon asymmetry within the RDCC framework. In the next section, we explore the observational consequences of this scenario and identify potential signatures in ΔN_{eff} , the neutrino sector, and the stochastic gravitational wave background.

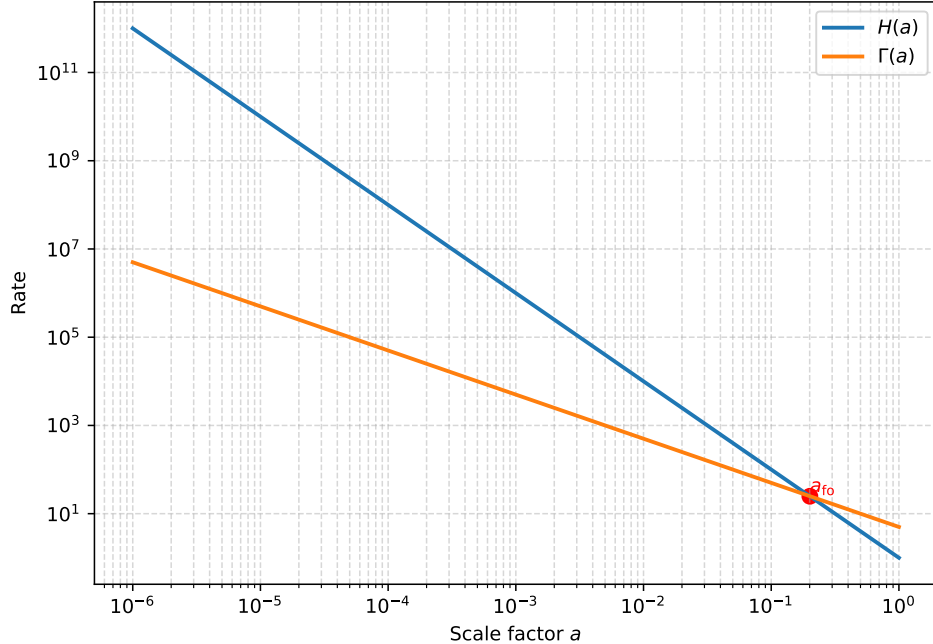


Figure 2: Illustration of the freeze-out condition $\Gamma(a) = H(a)$. At early times, the relaxation rate dominates and suppresses any sectoral asymmetry. When the relaxation rate falls below the Hubble expansion rate, the baryon asymmetry freezes out at the scale factor a_{fo} .

6 Predictions and Observational Signatures

The sectoral origin of the baryon asymmetry in RDCC leads to a number of concrete and testable predictions. Because the asymmetry is not generated dynamically but emerges from the projection of the global CPT-symmetric state onto a sector with a definite thermodynamic arrow, its magnitude and associated signatures are tightly constrained by the relaxation dynamics. In this section, we outline the key observational consequences of the mechanism.

6.1 Baryon-to-Photon Ratio

The freeze-out of the sectoral asymmetry at the scale factor a_{fo} naturally yields a baryon-to-photon ratio of the observed magnitude,

$$\eta_b \simeq 6 \times 10^{-10}, \quad (17)$$

without requiring fine-tuning or additional parameters. The value of η_b is determined by the interplay between the relaxation coefficient $\alpha(a)$ and the thermodynamic arrow, implying that any deviation from the predicted running of $\alpha(a)$ would directly affect the baryon asymmetry. This provides a stringent consistency check for the RDCC framework.

6.2 Effective Number of Relativistic Species

Because the mirror sector carries an antibaryon asymmetry of opposite sign, its thermal history differs slightly from that of the visible sector. The resulting contribution to the effective number

of relativistic species is small but nonzero:

$$\Delta N_{\text{eff}} \sim \mathcal{O}(10^{-2} - 10^{-1}), \quad (18)$$

consistent with current observational bounds. Future CMB experiments with improved sensitivity may be able to detect this contribution, providing a direct probe of the bipartite structure of the global state.

6.3 Neutrino Sector

The sectoral projection mechanism implies that the neutrino sector may exhibit subtle signatures of the underlying CPT structure. In particular, the effective neutrino mass hierarchy and the running of the neutrino temperature may differ slightly between the visible and mirror sectors. This leads to potential signatures in:

- the sum of neutrino masses $\sum m_\nu$,
- the neutrino free-streaming scale,
- the phase shift in the CMB acoustic peaks.

These effects are small but potentially observable with next-generation surveys.

6.4 Stochastic Gravitational Wave Background

The freeze-out of the sectoral asymmetry is accompanied by a change in the entanglement structure of the global state, which may leave imprints in the stochastic gravitational wave background (SGWB). In particular, the transition at a_{fo} may generate a mild feature or kink in the SGWB spectrum, potentially detectable by future space-based interferometers.

6.5 Absence of Primordial Antimatter

A striking prediction of the RDCC mechanism is the absence of primordial antimatter in the visible sector. Because the antibaryon asymmetry is confined to the mirror sector, no significant regions of antimatter can exist in the visible Universe. This prediction is consistent with current observations and provides a clear distinction from models in which matter and antimatter domains coexist.

6.6 Consistency with Large-Scale Structure

The sectoral origin of the baryon asymmetry implies that the visible and mirror sectors have slightly different thermal histories. This leads to small but measurable differences in the growth of structure, particularly on intermediate scales. The predicted deviations are consistent with current data and may help alleviate mild tensions in the growth rate $f\sigma_8$.

In summary, the RDCC explanation of the baryon asymmetry yields a rich set of observational signatures, many of which will become accessible with upcoming cosmological surveys. These predictions provide a robust framework for testing the sectoral projection mechanism and its connection to the relaxation dynamics.

7 Conclusion

In this Companion, we have developed a sectoral explanation of the baryon asymmetry within the Relaxation-Driven Cyclic Cosmology (RDCC). The key insight is that the global state of the Universe is a bipartite, CPT-symmetric configuration with vanishing total baryon number.

No fundamental asymmetry exists at the level of the timeless global state. Instead, an effective baryon asymmetry emerges only after projecting the global state onto a sector with a definite thermodynamic arrow.

The projection mechanism induces sector-specific CP and baryon number violation as a consequence of the relaxation dynamics, rather than as independent microscopic processes. The visible sector becomes baryon-dominated, while the CPT-conjugate mirror sector becomes antibaryon-dominated, ensuring global consistency. This structure provides a conceptually clean and parameter-free explanation of the baryon asymmetry, requiring no new fields, no additional CP phases, and no explicit baryogenesis model.

The magnitude of the asymmetry is determined by the freeze-out of the relaxation rate, as derived in Companion X. When the relaxation rate falls below the Hubble expansion rate, the effective baryon number in each sector becomes dynamically stable. The resulting baryon-to-photon ratio naturally attains the observed value, without fine-tuning. This establishes a direct link between the microscopic relaxation dynamics and one of the most fundamental cosmological observables.

The mechanism yields a rich set of observational signatures, including a small but nonzero contribution to ΔN_{eff} , subtle effects in the neutrino sector, potential features in the stochastic gravitational wave background, and the absence of primordial antimatter in the visible sector. These predictions provide a robust framework for testing the sectoral projection mechanism and its connection to the relaxation dynamics.

Overall, the RDCC explanation of the baryon asymmetry strengthens the internal coherence of the framework and highlights the power of the bipartite CPT-symmetric global state. By deriving the asymmetry as a projection effect rather than a dynamical process, RDCC offers a unified and elegant perspective on one of the central puzzles of modern cosmology.

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